

**M.Tech/PHD**

**END TERM EXAMINATION**

**May-2025**

**Pattern Recognition (AI504/ RICV502/RIRA502)**

**Time: 3:00 Hours**

**Max. Marks: 40**

**Note: Attempt any five questions including Q. No 1.**

Q.1 a) Explain the concept of variance in the context of Principal Component Analysis (PCA). Why does PCA maximize variance?

[BTL2][CO2][2M]

b) What are the advantages and disadvantages of using a higher value of  $k$  (e.g., 10 or more) in  $k$ -fold cross-validation?

[BTL4] [CO1][2M]

c) Elaborate the process of inference in Bayesian Networks.

[BTL2][CO3][2M]

d) Differentiate between SIR and LTM information diffusion model?

[BTL2][CO5][2M]

Q.2) Define **Independent Cascade Model (ICM)** of information diffusion in brief. In a social network, each individual has a 50% probability of influencing their neighbor to adopt a new product. Suppose an individual starts adopting the product (initially infected node), and they have 5 neighbors. Use the Independent Cascade Model (ICM) to calculate the expected number of individuals who will adopt the product after 3 steps. Assume that each adopter has 5 neighbors to influence, and that individuals who have already adopted the product cannot adopt again.

[BTL5][CO5]

[2+6=8M]

Q.3 Mathematically formulate node classification problem in an attributed complex Networks. Consider the following graph where each node has an initial label. The nodes have neighbors as described below:

- **Node 1:** Label = A, Neighbors = [Node 2, Node 3]
- **Node 2:** Label = B, Neighbors = [Node 1, Node 4]
- **Node 3:** Label = A, Neighbors = [Node 1, Node 5]



- **Node 4:** Label = ?, Neighbors = [Node 2, Node 5]
- **Node 5:** Label = ?, Neighbors = [Node 3, Node 4]

Apply the **Label Propagation Algorithm (LPA)** for 2 iterations and find the final labels of all nodes. The rule for label propagation is that in each iteration, each node adopts the label that is most frequent among its neighbors.

[BTL5][CO4][2+6=8M]

Q4. Explain hierarchical agglomerative clustering in brief. Consider the following dataset of 5 data points in a 2-dimensional feature space:

| Point | X1 | X2 |
|-------|----|----|
| A     | 1  | 2  |
| B     | 2  | 3  |
| C     | 6  | 8  |
| D     | 7  | 9  |
| E     | 3  | 4  |

You need to perform **hierarchical agglomerative clustering** using the **Euclidean distance** and the **single linkage** criterion (minimum distance between clusters).

Perform following steps.

**Step 1:** Calculate the pairwise Euclidean distances between the points.

**Step 2:** Perform hierarchical clustering using the single linkage method. Construct the **dendrogram** and determine how the clusters merge in each step.

**Step 3:** If you were to cut the dendrogram at a distance threshold of 5, how many clusters would you end up with, and what are the points in each cluster?

[BTL5][CO3][1+7=8M]

Q5. Explain the concept of neighborhood aggregation and message passing techniques used in Graph Neural Networks (GNNs). Describe the neighborhood aggregation process in GCN, GAT and GraphSAGE with suitable mathematical representation.

[BTL4][CO5][2+6=8M]

6. (a) Explain how Linear Discriminant Analysis (LDA) can be used for classification problems. A teacher wants to classify students as "Pass" or "Fail" based on their test scores using Linear Discriminant Analysis (LDA). The scores of previously evaluated students are as follows:

- **Fail:** 50, 60
- **Pass:** 70, 80, 90

You are given a new student with a score of 65. Use LDA to determine whether the student is classified as **Pass** or **Fail**.

[BTL5][CO3][1+4=5M]

(b) Consider the following set of data points in a 2-dimensional space:

(2,1),(2,3),(4,7),(5,8),(9,1), and (9,3)

Suppose we want to cluster these data points into  $k=2$  clusters using the k-means algorithm with the initial centroids chosen as (3,2) and (8,2). Perform one iteration of the k-means algorithm and find the coordinates of the two new centroids.

[BTL3] [CO2] [3M]